

Global Strategy for the Diagnosis, Management, and Prevention of Chronic Obstructive Lung Disease 2017 Report GOLD Executive Summary

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Abstract

This Executive Summary of the Global Strategy for the Diagnosis, Management, and Prevention of COPD, Global Initiative for Chronic Obstructive Lung Disease (GOLD) 2017 report focuses primarily on the revised and novel parts of the document. The most significant changes include: (1) the assessment of chronic obstructive pulmonary disease has been refined to separate the spirometric assessment from symptom evaluation. ABCD groups are now proposed to be derived exclusively from patient symptoms and their

history of exacerbations; (2) for each of the groups A to D, escalation strategies for pharmacologic treatments are proposed; (3) the concept of deescalation of therapy is introduced in the treatment assessment scheme; (4) nonpharmacologic therapies are comprehensively presented; and (5) the importance of comorbid conditions in managing chronic obstructive pulmonary disease is reviewed.

Keywords: chronic obstructive pulmonary disease; COPD diagnosis; COPD management; COPD prevention

Introduction

This Executive Summary of the Global Strategy for the Diagnosis, Management, and Prevention of Chronic Obstructive

Pulmonary Disease (COPD) Global Initiative for Chronic Obstructive Lung Disease (GOLD) 2017 report is based on peer-reviewed publications to October 2016. Levels of evidence are assigned to

evidence-based recommendations where appropriate. Categories used to grade the levels of evidence are provided in Table E1 in the online supplement.

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Influence Chronic Obstructive
Pulmonary Disease
Development and Progression****Key Points:**

- Chronic obstructive pulmonary disease (COPD) is a common, preventable, and treatable disease that is characterized by persistent respiratory symptoms and airflow limitation due to airway and/or alveolar abnormalities, usually caused by significant exposure to noxious particles or gases.
- Dyspnea, cough, and/or sputum production are the most frequent symptoms; symptoms are commonly underreported by patients.
- Tobacco smoking is the main risk exposure for COPD, but environmental exposures like biomass fuel exposure and air pollution may contribute. Besides exposures, host factors (genetic abnormalities, abnormal lung development, and accelerated aging)

predispose individuals to develop COPD.

- COPD may be punctuated by acute worsening of respiratory symptoms, called exacerbations.
- In most patients, COPD is associated with significant concomitant chronic diseases, which increase morbidity and mortality.

Definition and Pathogenesis

Chronic obstructive pulmonary disease (COPD) is a common, preventable, and treatable disease characterized by persistent respiratory symptoms and airflow limitation that is due to airway and/or alveolar abnormalities, usually caused by significant exposure to noxious particles or gases.

The chronic airflow limitation that characterizes COPD is caused by a mixture of small airway disease (e.g., obstructive bronchiolitis) and parenchymal destruction (emphysema), the relative contributions of which vary from person to person. Chronic inflammation causes structural changes, small airway narrowing, and destruction of

lung parenchyma. A loss of small airways may contribute to airflow limitation and mucociliary dysfunction, a characteristic feature of the disease.

Chronic respiratory symptoms may precede the development of airflow limitation and be associated with acute respiratory events (1). Chronic respiratory symptoms may exist in individuals with normal spirometry (1, 2), and a significant number of smokers without airflow limitation have structural evidence of lung disease manifested by the presence of emphysema, airway wall thickening, and gas trapping (1, 2).

**Factors That Influence Disease
Development and Progression**

Although cigarette smoking is the most well-studied COPD risk factor, epidemiologic studies demonstrate that nonsmokers may also develop chronic airflow limitation (3). Compared with smokers with COPD, never smokers with chronic airflow limitation have fewer symptoms, milder disease, and a lower burden of systemic inflammation (4). Never smokers with chronic airflow limitation do not have an increased risk of lung cancer or cardiovascular comorbidities; however, they have an

increased risk of pneumonia and mortality from respiratory failure (4).

Processes occurring during gestation, birth, and exposures during childhood and adolescence affect lung growth (5, 6). Reduced maximal attained lung function (as measured by spirometry) may identify individuals at increased risk for COPD (2, 7). Factors in early life termed “childhood disadvantage factors” are as important as heavy smoking in predicting lung function in adult life (8). An examination of three different longitudinal cohorts found that approximately 50% of patients developed COPD owing to an accelerated decline in FEV₁; the other 50% developed COPD owing to abnormal lung growth and development.

Cigarette smokers have a higher prevalence of respiratory symptoms and lung function abnormalities, a greater annual rate of decline in FEV₁, and a greater COPD mortality rate than nonsmokers (9). Other types of tobacco (e.g., pipe, cigar, water pipe) (10–12) and marijuana (13) are also risk factors for COPD. Passive exposure to cigarette smoke, also known as environmental tobacco smoke, may also contribute to respiratory symptoms and COPD (14) by increasing the lung’s total burden of inhaled particles and gases. Smoking during pregnancy may pose a risk for the fetus by affecting *in utero* lung growth and development and possibly priming the immune system (15).

Occupational exposures, including organic and inorganic dusts, chemical agents, and fumes, are underappreciated risk factors for COPD development (16, 17).

Wood, animal dung, crop residues, and coal, typically burned in open fires or poorly functioning stoves, may lead to indoor air pollution (18). Indoor pollution from biomass cooking and heating in poorly ventilated dwellings is a risk for COPD (19–21).

Asthma may be a risk for the development of chronic airflow limitation and COPD (22).

Airway hyperresponsiveness can exist without a clinical diagnosis of asthma and is an independent predictor of COPD and respiratory mortality in population studies (23, 24). It may indicate a risk for excessive lung function decline in mild COPD (25).

A history of severe childhood respiratory infection is associated with reduced lung

function and increased respiratory symptoms in adulthood (26). HIV infection accelerates the onset of smoking-related emphysema and COPD (27); tuberculosis has also been identified as a risk for COPD as well as a potential comorbidity (28–30).

Diagnosis and Initial Assessment

Key Points:

- COPD should be considered in any patient with dyspnea, chronic cough or sputum production, and/or a history of exposure to risk factors.
- Spirometry is required to make the diagnosis; a post-bronchodilator FEV₁/FVC less than 0.70 confirms the presence of persistent airflow limitation.
- The goals of COPD assessment are to determine the level of airflow limitation, the impact of disease on the patient’s health status, and the risk of future events (such as exacerbations, hospital admissions, or death) to guide therapy.
- Concomitant chronic diseases occur frequently in patients with COPD and should be treated because they can independently affect mortality and hospitalizations.

Diagnosis

COPD should be considered in any patient with dyspnea, chronic cough, or sputum production, and/or a history of exposure to risk factors for the disease (Figure 1 and Table 1). Spirometry is required to make the diagnosis in this clinical context (31); a post-bronchodilator FEV₁/FVC less than 0.70 confirms the presence of persistent airflow limitation and identifies the presence of COPD in patients with appropriate symptoms and predisposing risks.

Symptoms

Chronic and progressive dyspnea is the most characteristic symptom of COPD.

Dyspnea. Dyspnea is a major cause of disability and anxiety in COPD (32). The terms used to describe dyspnea vary individually and culturally (33).

Cough. Chronic cough is often the first symptom of COPD and is frequently discounted by the patient as a consequence of smoking and/or environmental exposures.

Sputum production. Regular sputum production for 3 months or more in 2 consecutive years is the classic definition of chronic bronchitis (34), an arbitrary definition that does not reflect the range of sputum production reported in COPD. Patients producing large volumes of sputum may have underlying bronchiectasis.

Wheezing and chest tightness.

Wheezing and chest tightness may vary between days and throughout a single day.

Additional features in severe disease.

Fatigue, weight loss, and anorexia are common in patients with more severe forms of COPD (35, 36).

Medical History

A detailed medical history of any patient who is known, or suspected, to have COPD should include:

- Exposure to risk factors, such as smoking and occupational or environmental exposures.
- Past medical history, including asthma, allergy, sinusitis, or nasal polyps; respiratory infections in childhood; other chronic respiratory and nonrespiratory diseases.
- Family history of COPD or other chronic respiratory diseases.
- Pattern of symptom development: age of onset, type of symptom, more frequent or prolonged “winter colds,” and social restriction.
- History of exacerbations or previous hospitalizations for a respiratory disorder.
- Presence of comorbidities, such as heart disease, osteoporosis, musculoskeletal disorders, and malignancies.
- Impact of disease on patient’s life, including limitation of activity, missed work and economic impact, and feelings of depression or anxiety.
- Social and family support available to the patient.
- Possibilities for reducing risk factors, especially smoking cessation.

Physical examination. Although important for general health, a physical examination is rarely diagnostic in COPD. Physical signs of airflow limitation/hyperinflation are usually not identifiable until significantly impaired lung function is present (37, 38).

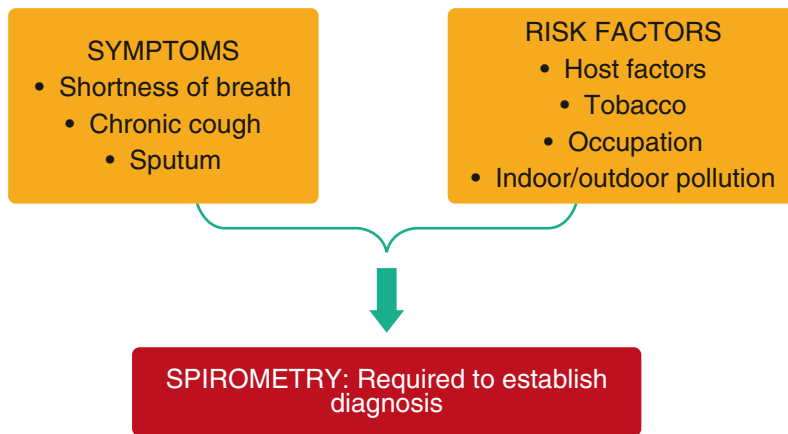


Figure 1. Pathways to the diagnosis of chronic obstructive pulmonary disease.

Spirometry. Spirometry is the most reproducible and objective measurement of airflow limitation. It is a noninvasive and readily available test. Good-quality spirometry is possible in any healthcare setting; all healthcare workers who care for patients with COPD should have access to spirometry.

A post-bronchodilator fixed ratio of FEV_1/FVC less than 0.70 is the spirometric criterion for airflow limitation. This criterion is simple and independent of reference values and has been used in numerous clinical trials. However, it may result in more frequent diagnosis of COPD in the elderly (39, 40) and less frequent diagnosis in adults younger than 45 years

(40), especially in mild disease, compared with a cutoff based on the lower limit of normal (LLN) values for FEV_1/FVC . Several limitations occur with using LLN as the diagnostic criterion for spirometric obstruction: (1) LLN values are dependent on the choice of reference equations that use post-bronchodilator FEV_1 , (2) there are no longitudinal studies that validate using the LLN, and (3) studies using LLN in populations where smoking is not the major cause of COPD are lacking.

Normal spirometry may be defined by a new approach from the Global Lung Initiative (GLI) (41, 42). Using GLI equations, z scores were calculated for FEV_1 , FVC, and FEV_1/FVC and compared

with fixed ratio data. The findings suggest that among adults with GLI-defined normal spirometry, the use of a fixed ratio may misclassify individuals as having respiratory impairment. These findings await additional study in other cohorts.

The risk of misdiagnosis and overtreatment using the fixed ratio as a diagnostic criterion is limited because spirometry is only one parameter used to establish the clinical diagnosis of COPD. GOLD favors using the fixed ratio over LLN because diagnostic simplicity and consistency are crucial for the busy clinician.

Assessing the degree of reversibility of airflow limitation (e.g., measuring FEV_1 before and after bronchodilator or corticosteroids) to make therapeutic decisions is not recommended (43) because it does not aid the diagnosis of COPD, differentiate COPD from asthma, or predict the long-term response to treatment (44).

In asymptomatic individuals without exposures to tobacco or other noxious stimuli, screening spirometry is not indicated. However, in those with symptoms and/or risk factors (e.g., >20 pack-years of smoking or recurrent chest infections), the diagnostic yield for COPD is relatively high and spirometry should be considered (45, 46). GOLD advocates active case finding (45, 47) (i.e., performing spirometry in patients with symptoms and/or risk factors) but not routine screening spirometry in asymptomatic individuals without COPD risk factors.

Table 1. Key Indicators for Considering a Diagnosis of Chronic Obstructive Pulmonary Disease

Consider COPD, and perform spirometry, if any of these indicators are present in an individual older than age 40 years. These indicators are not diagnostic themselves, but the presence of multiple key indicators increases the probability of a diagnosis of COPD. Spirometry is required to establish a diagnosis of COPD.

Dyspnea that is	Progressive over time Characteristically worse with exercise Persistent
Chronic cough	May be intermittent and may be unproductive Recurrent wheeze
Chronic sputum production	With any pattern
Recurrent lower respiratory tract infections	
History of risk factors	Host factors (such as genetic factors, congenital/developmental abnormalities, etc.) Tobacco smoke Smoke from home cooking and heating fuels Occupational dusts, vapors, fumes, gases and other chemicals
Family history of COPD and/or childhood factors	For example, low birthweight, childhood respiratory infections

Definition of abbreviation: COPD = chronic obstructive pulmonary disease.

Assessment

The goals of COPD assessment to guide therapy are (1) to determine the level of airflow limitation, (2) to define its impact on the patient's health status, and (3) to identify the risk of future events (such as exacerbations, hospital admissions, or death).

To achieve these goals, COPD assessment must consider separately the following aspects of the disease:

- Presence and severity of the spirometric abnormality
- Current nature and magnitude of symptoms
- History/future risk of exacerbations
- Presence of comorbidities

Classification of severity of airflow limitation. Spirometry should be performed after administration of an adequate dose of at least one short-acting inhaled bronchodilator to minimize variability.

The role of spirometry for the diagnosis, assessment, and follow-up of COPD is summarized in Table 2.

Assessment of Symptoms. COPD was previously viewed as a disease largely characterized by breathlessness. A simple measure of breathlessness, such as the modified British Medical Research Council (mMRC) Questionnaire (48), was considered adequate for assessment of symptoms (49–51). However, COPD impacts patients well beyond dyspnea (52). For this reason, a comprehensive assessment of symptoms is recommended. The most comprehensive disease-specific health status questionnaires include the Chronic Respiratory Questionnaire (53) and St. George's Respiratory Questionnaire (SGRQ) (54). These are too complex to use in clinical practice, but shorter measures, such as the COPD Assessment Test (CAT™), are suitable.

Choice of Thresholds. SGRQ scores less than 25 are uncommon in patients with COPD (55), and scores greater than or equal to 25 are very uncommon in healthy persons (56, 57). The equivalent cutoff point for the CAT is 10 (58). An mMRC threshold of greater than or equal to 2 is used to separate “less breathlessness” from “more breathlessness.”

Assessment of Exacerbation Risk. The best predictor of frequent exacerbations (defined as two or more exacerbations per year) is a history of earlier treated events (59). Hospitalization for a COPD exacerbation has a poor prognosis and an increased risk of death (60).

BLOOD EOSINOPHIL COUNT. *Post hoc* analysis of two clinical trials in patients with COPD with an exacerbation history showed that higher blood eosinophil counts may predict increased exacerbation rates in patients treated with long-acting β -agonists (LABAs) (without inhaled corticosteroids [ICSs]) (61, 62). The treatment effect of ICS/LABA versus LABA on exacerbations was greater in patients with higher blood eosinophil counts. These findings suggest that blood eosinophil counts (1) are a biomarker of exacerbation risk in patients with a history of exacerbations, and (2) can predict the effects of ICSs on exacerbation prevention. Prospective trials are required to validate the use of blood eosinophil counts to predict ICS effects, to determine a cutoff threshold for blood eosinophils that predicts exacerbation risk, and to clarify blood eosinophil cutoff values that could be used in clinical practice.

Table 2. Role of Spirometry

- Diagnosis
- Assessment of severity of airflow obstruction (for prognosis)
- Follow-up assessment
 - Therapeutic decisions
 - Pharmacologic in selected circumstances (e.g., discrepancy between spirometry and level of symptoms)
 - Consider alternative diagnoses when symptoms are disproportionate to degree of airflow obstruction
 - Nonpharmacologic (e.g., interventional procedures)
 - Identification of rapid decline

Assessment of Concomitant Chronic Diseases (Comorbidities). Patients with COPD often have important concomitant chronic illnesses as COPD represents an important component of multimorbidity, particularly in the elderly (60, 63–65).

Revised Combined COPD Assessment. The “ABCD” assessment tool of the 2011 GOLD Report was a major step forward from the simple spirometric grading system of earlier GOLD reports because it incorporated patient-reported outcomes and highlighted the importance of exacerbation prevention in COPD management. However, there were important limitations. ABCD assessment performed no better than spirometric grades for mortality prediction or other important health outcomes (66–68). Moreover, group D outcomes were modified by two parameters: lung function and/or exacerbation history, which caused

confusion (69). To address these concerns, the 2017 GOLD Report provides a refinement of the ABCD assessment that separates spirometric grades from ABCD groupings. For some therapy recommendations, especially pharmacologic treatments, ABCD groups are derived exclusively from patient symptoms and their exacerbation history. However, spirometry, in conjunction with patient symptoms and exacerbation history, remains vital for the diagnosis, prognostication, and consideration of other important therapeutic approaches, especially nonpharmacologic therapies. This new approach to assessment is illustrated in Figure 2.

In the refined assessment scheme, patients should undergo spirometry to determine the severity of airflow limitation (i.e., spirometric grade). They should also undergo assessment of either dyspnea using

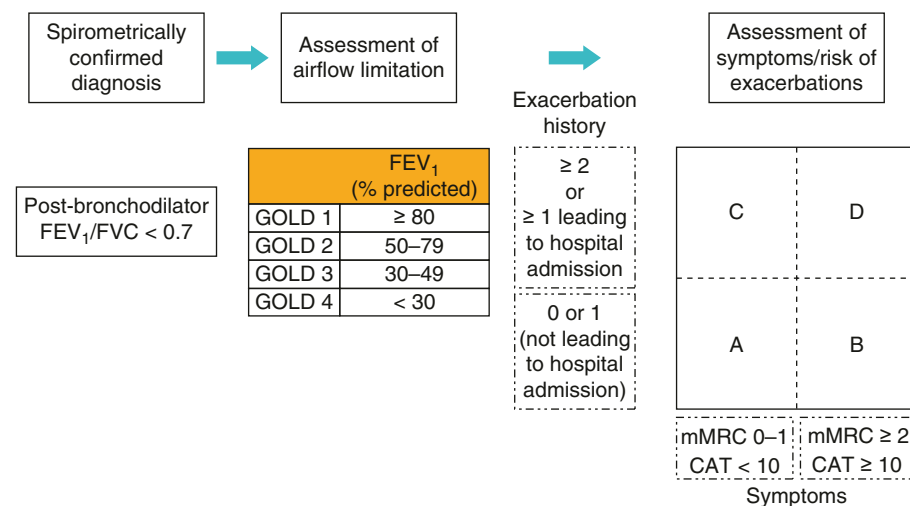


Figure 2. The refined ABCD assessment tool. CAT = COPD Assessment Test; COPD = chronic obstructive pulmonary disease; GOLD = Global Initiative for Chronic Obstructive Lung Disease; mMRC = modified British Medical Research Council questionnaire.

mMRC or symptoms using CAT. Finally, their history of exacerbations (including prior hospitalizations) should be recorded.

The number provides information regarding severity of airflow limitation (spirometric grades 1 to 4), whereas the letter (groups A to D) provides information regarding symptom burden and risk of exacerbation. FEV₁ is a very important parameter at the population level in the prediction of important clinical outcomes, such as mortality and hospitalizations, or prompting consideration for nonpharmacologic therapies, such as lung reduction or lung transplantation. However, at the individual patient level, FEV₁ loses precision and thus cannot be used alone to determine all therapeutic options. Furthermore, in some circumstances, such as during hospitalization or urgent presentation to the clinic or emergency room, the ability to assess patients on the basis of symptoms and exacerbation history, independent of the spirometric value, allows clinicians to initiate a treatment plan on the basis of the revised ABCD scheme. This approach acknowledges the limitations of FEV₁ in making treatment decisions for individualized patient care and highlights the importance of patient symptoms and exacerbation risks in guiding therapies in COPD. The separation of airflow limitation from clinical parameters makes it clearer what is being evaluated and ranked. This should facilitate more precise treatment recommendations on the basis of parameters that are driving the patient's symptoms at any given time.

EXAMPLE. Consider two patients—both patients with FEV₁ less than 30% predicted, CAT scores of 18, one with no exacerbations in the past year and the other with three exacerbations in the past year. Both would have been labeled GOLD D in the prior classification scheme. However, with the new proposed scheme, the subject with three exacerbations in the past year would be labeled GOLD grade 4, group D. Individual decisions on pharmacotherapeutic approaches would use the recommendations based on the ABCD assessment to treat the patient's major problem at this time (i.e., persistent exacerbations). The other patient, who has had no exacerbations, would be classified as GOLD grade 4, group B. In such patients—besides pharmacotherapy and rehabilitation—lung reduction, lung transplantation, or bullectomy may be

important therapeutic considerations, given their symptom burden and level of spirometric limitation.

Alpha-1 Antitrypsin Deficiency. The World Health Organization recommends that all patients with a diagnosis of COPD be screened once for alpha-1 antitrypsin deficiency (70). A low concentration (<20% of normal) is suggestive of homozygous deficiency. Family members should be screened and together with the patient referred to specialist centers for advice and management.

Additional Investigations

To rule out other concomitant disease contributing to respiratory symptoms, or in cases where patients do not respond to the treatment plan as expected, additional testing may be required. Thoracic imaging (chest X-ray, chest computed tomography [CT]), assessment of lung volumes and/or diffusion capacity, oximetry and arterial blood gas measurement, and exercise testing and assessment of physical activity should be considered.

COMPOSITE SCORES. The BODE (Body mass index, Obstruction, Dyspnea, and Exercise) method gives a composite score that is a better predictor of subsequent survival than any single component (71). Simpler alternatives that do not include exercise testing need validation to confirm suitability for routine clinical use (72, 73).

DIFFERENTIAL DIAGNOSES. In some patients, features of asthma and COPD may coexist. The terms asthma-COPD overlap syndrome (ACOS) or asthma-COPD overlap (ACO) acknowledge the overlap of these two common disorders causing chronic airflow limitation rather than designating it a distinct syndrome. Most other potential differential diagnoses are easier to distinguish from COPD.

OTHER CONSIDERATIONS. Some patients without evidence of airflow limitation have evidence of structural lung disease on chest imaging (emphysema, gas trapping, airway wall thickening). Such patients may report exacerbations of respiratory symptoms or even require treatment with respiratory medications on a chronic basis. Whether these patients have acute or chronic bronchitis, a persistent form of asthma, or an earlier presentation of what will become COPD as it is currently defined is unclear and requires further study.

Prevention and Maintenance Therapy

Key Points:

- Smoking cessation is key. Pharmacotherapy and nicotine replacement increase long-term smoking abstinence rates.
- The effectiveness and safety of e-cigarettes as a smoking cessation aid is uncertain.
- Pharmacologic therapy can reduce COPD symptoms, reduce the frequency and severity of exacerbations, and improve health status and exercise tolerance.
- Each pharmacologic treatment regimen should be individualized and guided by the severity of symptoms, risk of exacerbations, side effects, comorbidities, drug availability and cost, and the patient's response, preference, and ability to use various drug delivery devices.
- Inhaler technique needs to be assessed regularly.
- Influenza and pneumococcal vaccinations decrease the incidence of lower respiratory tract infections.
- Pulmonary rehabilitation improves symptoms, quality of life, and physical and emotional participation in everyday activities.
- In patients with severe resting chronic hypoxemia, long-term oxygen therapy improves survival.
- In patients with stable COPD and resting or exercise-induced moderate desaturation, long-term oxygen treatment should not be prescribed routinely; however, individual patient factors should be considered.
- In patients with severe chronic hypercapnia and a history of hospitalization for acute respiratory failure, long-term noninvasive ventilation may decrease mortality and prevent rehospitalization.
- In select patients with advanced emphysema refractory to optimized medical care, surgical or bronchoscopic interventional treatments may be beneficial.
- Palliative approaches are effective in controlling symptoms in advanced COPD.

Smoking Cessation

Smoking cessation influences the natural history of COPD. If effective resources and time are dedicated to smoking cessation, long-term quit success rates of up to 25% can be achieved (74).

Nicotine replacement products.

Nicotine replacement therapy increases long-term smoking abstinence rates (75–77) and is more effective than placebo. E-cigarettes are increasingly used as a form of nicotine replacement therapy, although their efficacy remains controversial (78–82).

Pharmacologic products. Varenicline (83), bupropion (84), and nortriptyline (85) increase long-term quit rates (85) but should be used as part of an interventional program rather than as a sole intervention.

Smoking cessation programs. A five-step program for intervention (75) provides a framework to guide healthcare providers to help patients stop smoking (75, 77, 86). Counseling delivered by health professionals significantly increases quit rates over self-initiated strategies (87). The combination of pharmacotherapy and behavioral support increases smoking cessation rates (88).

Vaccinations

Influenza vaccine and pneumococcal vaccines.

Influenza vaccination reduces serious illness (89), death (90–93), the risk of ischemic heart disease (94), and the total number of exacerbations (90). Vaccines containing either killed or live inactivated viruses are recommended (95) as they are more effective in elderly patients with COPD (96).

The pneumococcal vaccinations pneumococcal conjugate vaccine (PCV13) and pneumococcal polysaccharide vaccine (PPSV23) are recommended for all patients 65 years of age and older (Table E2).

Pharmacologic Therapy for Stable COPD

Overview of medications. Pharmacologic therapy for COPD reduces symptoms and the frequency and severity of exacerbations and improves exercise tolerance and health status. No existing medication modifies the long-term decline in lung function (97–101). The classes of medications used to treat COPD are shown in Table E3. The choice within each class depends on the availability and cost of medication and

favorable clinical response balanced against side effects. Each treatment regimen needs to be individualized as the relationship between severity of symptoms, airflow limitation, and severity of exacerbations varies between patients.

Bronchodilators. Bronchodilators increase FEV₁, reduce dynamic hyperinflation at rest and during exercise (102, 103), and improve exercise performance. Bronchodilator medications are usually given on a regular basis to prevent or reduce symptoms. Toxicity is dose related.

β₂-AGONISTS. β₂-agonists, including short-acting β₂-agonist (SABA) and LABA agents, relax airway smooth muscle. Stimulation of β₂-adrenergic receptors can produce resting sinus tachycardia and precipitate cardiac rhythm disturbances in susceptible patients. Exaggerated somatic tremor occurs in some patients treated with higher doses of β₂-agonists.

ANTIMUSCARINIC DRUGS.

Ipratropium, a short-acting muscarinic antagonist, provides small benefits over SABAs in terms of lung function, health status, and requirement for oral steroids (104). Long-acting muscarinic antagonist (LAMA) treatment improves symptoms and health status (105, 106), improves the effectiveness of pulmonary rehabilitation (107, 108), and reduces exacerbations and related hospitalizations (107). Clinical trials

have shown a greater effect on exacerbation rates for LAMA treatment (tiotropium) versus LABA treatment (109, 110). An unexpected small increase in cardiovascular events was reported in patients with COPD regularly treated with ipratropium bromide (111, 112). A large trial reported no difference in mortality, cardiovascular morbidity, or exacerbation rates when using tiotropium as a dry-powder inhaler compared with a mist delivered by the Respimat inhaler (113).

METHYLXANTHINES. Theophylline exerts a modest bronchodilator effect in stable COPD (114) and improves FEV₁ and breathlessness when added to salmeterol (115, 116). There is limited and contradictory evidence regarding the effect of low-dose theophylline on exacerbation rates (117, 118). Toxicity is dose related, which is a problem as most of the benefit occurs when near-toxic doses are given (114, 119).

Combination bronchodilator therapy.

Combining bronchodilators with different mechanisms and durations of action may increase the degree of bronchodilation, with a lower risk of side effects compared with increasing the dose of a single bronchodilator (Table 3) (120). There are numerous combinations of LABAs and LAMAs in a single inhaler available (Table E3). These combinations improve lung function compared with placebo (120) and have a greater impact on patient-reported

Table 3. Bronchodilators in Stable Chronic Obstructive Pulmonary Disease

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- Inhaled bronchodilators in COPD are central to symptom management and commonly given on a regular basis to prevent or reduce symptoms (**Evidence A**).
 - Regular and as-needed use of SABA or SAMA improves FEV₁ and symptoms (**Evidence A**).
 - Combinations of SABA and SAMA are superior to either medication alone in improving FEV₁ and symptoms (**Evidence A**).
 - LABAs and LAMAs significantly improve lung function, dyspnea, and health status and reduce exacerbation rates (**Evidence A**).
 - LAMAs have a greater effect on exacerbation reduction compared with LABAs (**Evidence A**) and decrease hospitalizations (**Evidence B**).
 - Combination treatment with LABA and LAMA increases FEV₁ and reduces symptoms compared with monotherapy (**Evidence A**).
 - Combination treatment with LABA and LAMA reduces exacerbations compared with monotherapy (**Evidence B**) or ICS/LABA (**Evidence B**).
 - Tiotropium improves the effectiveness of pulmonary rehabilitation in increasing exercise performance (**Evidence B**).
 - Theophylline exerts a small bronchodilator effect in stable COPD (**Evidence A**) that is associated with modest symptomatic benefits (**Evidence B**).
-

Definition of abbreviations: COPD = chronic obstructive pulmonary disease; ICS = inhaled corticosteroid; LABA = long-acting β-agonist; LAMA = long-acting muscarinic antagonist; SABA = short-acting β-agonist; SAMA = short-acting muscarinic antagonist.

outcomes than monotherapies (121–124). LABA/LAMA combination improves symptoms and health status in patients with COPD (125), is more effective than long-acting bronchodilator monotherapy for preventing exacerbations (126), and decreases exacerbations to a greater extent than ICS/LABA combination (127).

Antiinflammatory agents.

Exacerbations represent the main clinically relevant end point used for the efficacy assessment of antiinflammatory drugs (Table 4).

ICSs. In patients with moderate to very severe COPD and exacerbations, an ICS combined with an LABA is more effective than either component alone in improving lung function and health status and reducing exacerbations (128, 129).

However, survival is not affected by combination therapy (130, 131).

ICS use has a higher prevalence of oral candidiasis, hoarse voice, skin bruising, and pneumonia (132). Patients at higher risk of pneumonia include those who currently smoke, are aged 55 years or older, have a history of prior exacerbations or pneumonia, have a body mass index (BMI) less than 25 kg/m², or have a poor MRC dyspnea grade and/or severe airflow limitation (133).

Results from randomized controlled trials (RCTs) have yielded variable results regarding the risk of decreased bone density and fractures with ICS treatment (99, 134–137). Observational studies suggest that ICS treatment could be associated with increased risks of diabetes/poor control of

diabetes (138), cataracts (139), and mycobacterial infection (140), including tuberculosis (141, 142).

ICS WITHDRAWAL. Withdrawal studies provide equivocal results regarding the consequences of withdrawal on lung function, symptoms, and exacerbations (143–147).

Triple inhaled therapy. Combination of LABA plus LAMA plus ICS (triple therapy) may improve lung function and patient-reported outcomes (148–151) and reduce exacerbation risk (149, 152–154). However, one RCT failed to demonstrate any benefit of adding an ICS to LABA plus LAMA on exacerbations (155). More evidence is needed to compare the benefits of triple therapy (LABA/LAMA/ICS) to LABA/LAMA.

Oral glucocorticoids. Oral glucocorticoids have no role in the chronic daily treatment of COPD because of a lack of benefit balanced against a high rate of systemic complications.

Phosphodiesterase-4 inhibitors.

Roflumilast reduces moderate and severe exacerbations treated with systemic corticosteroids in patients with chronic bronchitis, severe to very severe COPD, and a history of exacerbations (156). Phosphodiesterase-4 inhibitors have more adverse effects than inhaled medications for COPD (157). The most frequent are diarrhea, nausea, reduced appetite, weight loss, abdominal pain, sleep disturbance, and headache. Roflumilast should be avoided in underweight patients and used with caution in patients with depression.

Antibiotics. Azithromycin (250 mg/d or 500 mg three times per wk) or erythromycin (500 mg two times per d) for 1 year reduces the risk of exacerbations in patients prone to exacerbations (158–160). Azithromycin use showed a reduced exacerbation rate in former smokers only and was associated with an increased incidence of bacterial resistance and impaired hearing tests (160). Pulse moxifloxacin therapy in patients with chronic bronchitis and frequent exacerbations does not reduce exacerbation rate (161).

Mucolytic (mucokinetics, mucoregulators) and antioxidant agents (N-acetylcysteine, carbocysteine). Regular treatment with mucolytics, such as carbocysteine and N-acetylcysteine, may reduce exacerbations and modestly improve health status in patients not receiving ICSs (162, 163).

Table 4. Antiinflammatory Therapy in Stable Chronic Obstructive Pulmonary Disease

ICSs

- An ICS combined with an LABA is more effective than the individual components in improving lung function and health status and reducing exacerbations in patients with exacerbations and moderate to very severe COPD (**Evidence A**).
- Regular treatment with ICSs increases the risk of pneumonia, especially in those with severe disease (**Evidence A**).
- Triple inhaled therapy of ICS/LAMA/LABA improves lung function, symptoms, and health status (**Evidence A**) and reduces exacerbations (**Evidence B**) compared with ICS/LABA or LAMA monotherapy.

Oral glucocorticoids

- Long-term use of oral glucocorticoids has numerous side effects (**Evidence A**) with no evidence of benefits (**Evidence C**).

PDE4 inhibitors

- In patients with chronic bronchitis, severe to very severe COPD, and a history of exacerbations:
 - A PDE4 inhibitor improves lung function and reduces moderate and severe exacerbations (**Evidence A**).
 - A PDE4 inhibitor improves lung function and decreases exacerbations in patients who are on fixed-dose LABA/ICS combinations (**Evidence B**).

Antibiotics

- Long-term azithromycin and erythromycin therapy reduces exacerbations over 1 year (**Evidence A**).
- Treatment with azithromycin is associated with an increased incidence of bacterial resistance (**Evidence A**) and hearing test impairment (**Evidence B**).

Mucolytics/antioxidants

- Regular use of NAC and carbocysteine reduces the risk of exacerbations in select populations (**Evidence B**).

Other antiinflammatory agents

- Simvastatin does not prevent exacerbations in patients with COPD at increased risk of exacerbations and without indications for statin therapy (**Evidence A**). However, observational studies suggest that statins may have positive effects on some outcomes in patients with COPD who receive them for cardiovascular and metabolic indications (**Evidence C**).
- Leukotriene modifiers have not been tested adequately in patients with COPD.

Definition of abbreviations: COPD = chronic obstructive pulmonary disease; ICS = inhaled corticosteroid; LABA = long-acting β -agonist; LAMA = long-acting muscarinic antagonist; NAC = N-acetylcysteine; PDE4 = phosphodiesterase 4.

Other drugs with antiinflammatory potential. Although RCTs suggest that immunoregulators decrease the severity and frequency of exacerbations (164, 165), the long-term effects of this therapy are unknown. Nedocromil and leukotriene modifiers have not been adequately tested in COPD (166). There was no evidence of benefit, and some evidence of harm, after treatment with an anti-tumor necrosis factor- α antibody (infliximab) in moderate to severe COPD (167). Simvastatin did not prevent exacerbations in patients with COPD who had no metabolic or cardiovascular indication for statin treatment (168). An association between statin use and improved outcomes has been reported in observational studies of patients with COPD who received them for cardiovascular and metabolic indications (169). There is no evidence that vitamin D supplementation reduces exacerbations in unselected patients (170).

Issues related to inhaled delivery. Observational studies have identified a significant relationship between poor inhaler use and symptom control in COPD (171). Determinants of poor inhaler technique include older age, use of multiple devices, and lack of previous education on inhaler technique (172). Education improves inhalation technique in some but not all patients (172), especially when the “teach-back” approach is implemented (173).

Other pharmacologic treatments for COPD are summarized in Table E4.

ALPHA-1 ANTITRYPSIN AUGMENTATION THERAPY. Observational studies suggest a reduction in spirometric progression in alpha-1 antitrypsin deficiency patients treated with augmentation therapy versus patients who are not treated (174). Studies using sensitive parameters of emphysema progression determined by CT scans provide evidence for an effect on preserving lung tissue compared with placebo (175–177).

ANTITUSSIVES. The role of antitussives in patients with COPD is inconclusive (178).

VASODILATORS. Available studies report worsening gas exchange (179) with little improvement in exercise capacity or health status in patients with COPD (180, 181).

Rehabilitation, Education, and Self-Management

Pulmonary Rehabilitation

Pulmonary rehabilitation is a comprehensive intervention based on thorough patient

assessment followed by patient-tailored therapies (e.g., exercise training, education, and self-management interventions aimed at behavior changes to improve physical and psychological condition and promote adherence to health-enhancing behaviors in patients with COPD) (182). The benefits of pulmonary rehabilitation are considerable (Table E5). Pulmonary rehabilitation can reduce readmissions and mortality in patients after a recent exacerbation (≤ 4 wk from prior hospitalization) (183). Initiating pulmonary rehabilitation before hospital discharge, however, may compromise survival (184).

Pulmonary rehabilitation represents integrated patient management that includes a range of healthcare professionals (185) and sites, including hospital inpatient and outpatient settings and/or the patient's home (182).

Education, Self-Management, and Integrative Care

Education. Smoking cessation, correct use of inhaler devices, early recognition of exacerbation, decision-making, when to seek help, surgical interventions, and the consideration of advance directives are examples of educational topics.

Self-management. Self-management interventions that use written negotiated action plans for worsening symptoms may lead to less respiratory-related hospitalization and all-cause hospitalizations and improved health status (186). The health benefits of COPD self-management programs may be negated by increased mortality (187, 188). Generalization to real life remains difficult.

Integrated care programs. Integrated care programs improve several clinical outcomes, although not mortality (189). However, a large multicenter study within an existing well-organized system of care did not confirm this (190). Delivering integrated interventions by telemedicine provided no significant benefit (191).

Supportive, Palliative, End-of-Life, and Hospice Care

Symptom Control and Palliative Care

The goal of palliative care is to prevent and relieve suffering and to improve quality of life for patients and their families, regardless of the stage of disease or the need for other therapies (192). Palliation efforts should be

focused on the relief of dyspnea, pain, anxiety, depression, fatigue, and poor nutrition.

End-of-Life and Hospice Care

End-of-life care discussions should include patients and their families (193). Advance care planning can reduce anxiety for patients and their families; ensure that care is consistent with their wishes; and avoid unnecessary, unwanted, and costly invasive therapies (194, 195). Table E6 summarizes the approach to palliation, end-of-life, and hospice care.

Other Treatments

Oxygen Therapy and Ventilatory Support

Oxygen therapy. The long-term administration of oxygen (> 15 h/d) to patients with chronic respiratory failure increases survival in patients with severe resting hypoxemia (196). Long-term oxygen therapy does not lengthen time to death or first hospitalization or provide sustained benefit for any of the measured outcomes in patients with stable COPD and resting or exercise-induced moderate arterial oxygen desaturation (197).

Ventilatory support. Whether to use noninvasive positive pressure ventilation chronically at home to treat patients with acute-on-chronic respiratory failure after hospitalization remains undetermined. Retrospective studies have provided inconclusive data (198, 199). RCTs have yielded conflicting data on the use of home noninvasive positive pressure ventilation on survival and rehospitalization in chronic hypercapnic COPD (200–203). In patients with both COPD and obstructive sleep apnea (OSA), continuous positive airway pressure improves survival and avoids hospitalization (Table E7) (204).

Interventional Therapy

Surgical Interventions

Lung volume reduction surgery. An RCT confirmed that patients with COPD with upper-lobe emphysema and low postrehabilitation exercise capacity experienced improved survival when treated with lung volume reduction surgery (LVRS) compared with medical treatment (205). In patients with high postpulmonary rehabilitation exercise capacity, no difference in survival was noted after LVRS, although

health status and exercise capacity improved. LVRS has been demonstrated to result in higher mortality than medical management in patients with severe emphysema with an FEV₁ less than or equal to 20% predicted and either homogeneous emphysema in high-resolution CT (HRCT) or diffusing capacity of the lung for carbon monoxide of less than or equal to 20% predicted (206).

Bullectomy. In selected patients with relatively preserved underlying lung, bullectomy is associated with decreased dyspnea and improved lung function and exercise tolerance (207).

Lung transplantation. In selected patients, lung transplantation has been shown to improve health status and functional capacity but not to prolong survival (207–209). Bilateral lung transplantation has been reported to have longer survival than single lung transplantation in patients with COPD, especially those younger than 60 years of age (210).

Bronchoscopic Interventions to Reduce Hyperinflation in Severe Emphysema

Less-invasive bronchoscopic approaches to lung reduction have been developed (211). Prospective studies have shown that the use of bronchial stents is not effective (212), whereas use of lung sealant caused significant morbidity and mortality (213). An RCT of endobronchial valve placement showed statistically significant improvements in FEV₁ and 6-minute-walk distance compared with control therapy at 6 months postintervention (214), but the magnitude of the observed improvements was not clinically meaningful. Subsequently, efficacy of the same endobronchial valve has been studied in patients with heterogeneous (215) or heterogeneous and homogenous emphysema (216), with mixed outcomes.

Two multicenter trials that examined nitinol coils implanted into the lung compared with usual care reported increases in 6-minute-walk distance with coil treatment compared with control and smaller improvements in FEV₁ and quality of life measured by SGRQ (217, 218).

Additional data are needed to define the optimal patient population to receive a specific bronchoscopic lung volume technique and to compare the long-term durability of improvements in functional or physiological performance to LVRS relative to side effects (218).

Key points for interventional therapy in stable COPD are summarized in Table E8.

Management of Stable COPD

Key Points:

- The management strategy for stable COPD should be based on individualized symptom assessment and future risk of exacerbations.
- All individuals who smoke should be supported to quit.
- The main treatment goals are reduction of symptoms and future risk of exacerbations.
- Management strategies are not limited to pharmacologic treatments and should be complemented by appropriate nonpharmacologic interventions.

Effective COPD management should be based on an individualized assessment to reduce both current symptoms and future risk of exacerbations (Figure E1).

We propose personalization of initiating and escalating/deescalating treatments on the basis of the level of symptoms and an individual's risk of exacerbations. These recommendations are partially based on evidence generated in RCTs. These recommendations are intended to support clinician decision-making.

Identify and Reduce Exposure to Risk Factors

Cigarette smoking is the most commonly encountered and easily identifiable risk factor for COPD; smoking cessation should be continually encouraged for current

smokers. Reduction of total personal exposure to occupational dusts, fumes, and gases, and to indoor and outdoor air pollutants, should be addressed.

Treatment of Stable COPD

Pharmacologic Treatment

Pharmacologic therapies can reduce symptoms and the risk and severity of exacerbations and improve health status and exercise tolerance. The choice within each class depends on the availability of medication and the patient's response and preference (Tables 5–7).

Pharmacologic treatment algorithms. A proposed model for the initiation and then subsequent escalation and/or deescalation of pharmacologic management according to the individualized assessment of symptoms and exacerbation risk is shown in Figure 3. In past GOLD Reports, recommendations were only given for initial therapy. However, many patients with COPD are already receiving treatment and return with persistent symptoms after initial therapy or, less commonly, with resolution of some symptoms that may subsequently require less therapy. Therefore, we now suggest escalation and deescalation strategies. The recommendations are based on available efficacy and safety data. We acknowledge that treatment escalation has not been systematically tested; trials of deescalation are also limited and only include ICSs. There is a lack of direct evidence supporting the therapeutic recommendations for patients in groups C and D. These recommendations will be reevaluated as additional data become available.

GROUP A. All group A patients should be offered a bronchodilator to reduce breathlessness. This can be either a short- or a long-acting bronchodilator, depending on the individual patient's preference. The bronchodilator should

Table 5. Key Points for the Use of Bronchodilators

- LABAs and LAMAs are preferred over short-acting agents except for patients with only occasional dyspnea (**Evidence A**).
- Patients may be started on single long-acting bronchodilator therapy or dual long-acting bronchodilator therapy. In patients with persistent dyspnea on one bronchodilator, treatment should be escalated to two (**Evidence A**).
- Inhaled bronchodilators are recommended over oral bronchodilators (**Evidence A**).
- Theophylline is not recommended unless other long-term treatment bronchodilators are unavailable or unaffordable (**Evidence B**).

Definition of abbreviations: LABA = long-acting β -agonist; LAMA = long-acting muscarinic antagonist.

Table 6. Key Points for the Use of Antiinflammatory Agents

- Long-term monotherapy with ICSs is not recommended (**Evidence A**).
- Long-term treatment with ICSs may be considered in association with LABAs for patients with a history of exacerbations despite appropriate treatment with long-acting bronchodilators (**Evidence A**).
- Long-term therapy with oral corticosteroids is not recommended (**Evidence A**).
- In patients with exacerbations despite LABA/ICS or LABA/LAMA/ICS, chronic bronchitis, and severe to very severe airflow obstruction, the addition of a PDE4 inhibitor can be considered (**Evidence B**).
- In former smokers with exacerbations despite appropriate therapy, macrolides can be considered (**Evidence B**).
- Statin therapy is not recommended for prevention of exacerbations (**Evidence A**).
- Antioxidant mucolytics are recommended only in selected patients (**Evidence A**).

Definition of abbreviations: ICS = inhaled corticosteroid; LABA = long-acting β -agonist; LAMA = long-acting muscarinic antagonist; PDE4 = phosphodiesterase 4.

be continued if symptomatic benefit is noted.

GROUP B. Initial therapy should be a long-acting bronchodilator. Long-acting bronchodilators are superior to short-acting bronchodilators taken intermittently (104, 219). There is no evidence to recommend one class of long-acting bronchodilators over another for symptom relief; the choice should depend on individual patient response.

For patients with persistent breathlessness on monotherapy (220), the use of two bronchodilators is recommended. For patients with severe breathlessness, initial therapy with two bronchodilators may be considered.

GROUP C. Initial therapy should be a single long-acting bronchodilator. In two head-to-head comparisons (110, 111), the LAMA tested superior to the LABA regarding exacerbation prevention; therefore, we recommend initiating a LAMA in this group.

Patients with persistent exacerbations may benefit from adding a second long-acting bronchodilator (LABA/LAMA) or using a combination of an LABA and an ICS. As ICSs increase the risk for developing pneumonia, our primary choice is LABA/LAMA.

GROUP D. We recommend initiating an LABA/LAMA combination because:

- In studies with patient-reported outcomes as the primary endpoint, LABA/LAMA combinations showed superior results compared with a single bronchodilator.
- LABA/LAMA combination was superior to LABA/ICS combination in preventing exacerbations and improving other patient-reported outcomes in group D patients.
- Group D patients are at higher risk for pneumonia when receiving ICS treatment (111, 133).

If a single bronchodilator is initially chosen, an LAMA is preferred for exacerbation prevention on the basis of comparison to LABAs.

LABA/ICS may be the first choice for initial therapy in some patients. These patients may have a history and/or findings suggestive of ACO and/or high blood eosinophil counts.

In patients who develop additional exacerbations on LABA/LAMA therapy, we suggest two alternative pathways:

- Escalation to LABA/LAMA/ICS.
- Switch to LABA/ICS. If LABA/ICS therapy does not positively impact exacerbations/symptoms, an LAMA can be added.

If patients treated with LABA/LAMA/ICS still have exacerbations, the following options may be considered:

- Add roflumilast. This may be considered in patients with an FEV₁ less than 50% predicted and chronic bronchitis (221), particularly if they experienced at least one hospitalization for an exacerbation in the previous year (222).
- Add a macrolide in former smokers. The possibility of developing resistant organisms should be factored into the decision-making.
- Stopping ICS. This recommendation is supported by data that show an elevated risk of adverse effects (including pneumonia) and no significant harm from ICS withdrawal.

Nonpharmacologic Treatment

Education and self-management. An individual patient's evaluation and risk assessment (e.g., exacerbations, patient's needs, preferences, and personal goals) should aid the design of personalized self-management.

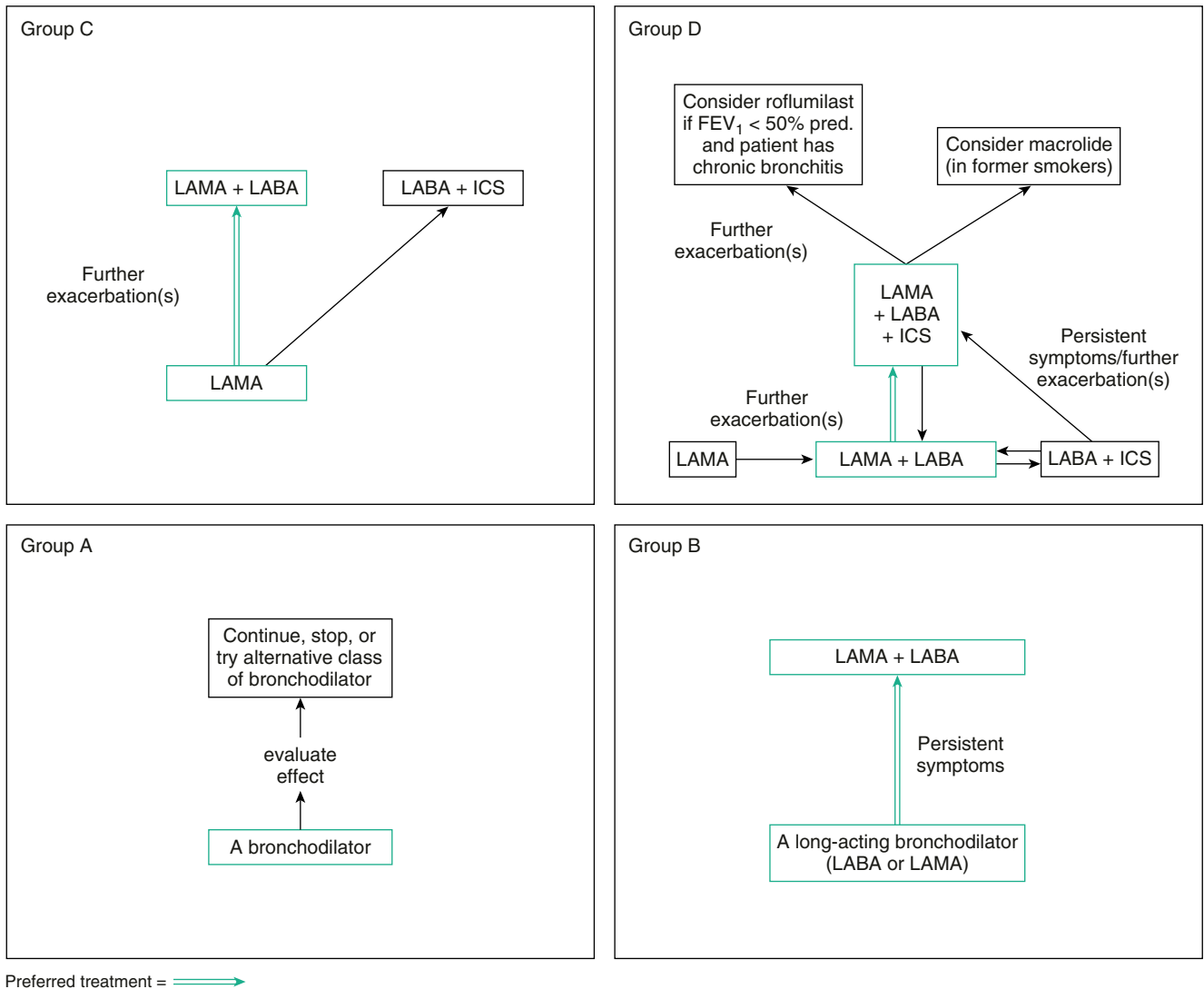
Pulmonary rehabilitation programs. Patients with high symptom burden and risk of exacerbations (groups B, C, and D), should take part in a full rehabilitation program that considers the individual's characteristics and comorbidities (182, 187, 223).

Exercise training. A combination of constant load or interval training with strength training provides better outcomes than either method alone (224). Adding strength training to aerobic training is effective in improving strength but does not improve health status or exercise tolerance (225). Upper extremity exercise training

Table 7. Key Points for the Use of Other Pharmacologic Treatments

- Patients with severe hereditary alpha-1 antitrypsin deficiency and established emphysema may be candidates for alpha-1 antitrypsin augmentation therapy (**Evidence B**).
- Antitussives cannot be recommended (**Evidence C**).
- Drugs approved for primary pulmonary hypertension are not recommended for patients with pulmonary hypertension secondary to COPD (**Evidence B**).
- Low-dose long-acting oral and parenteral opioids may be considered for treating dyspnea in patients with COPD with severe disease (**Evidence B**).

Definition of abbreviation: COPD = chronic obstructive pulmonary disease.



In patients with a major discrepancy between the perceived level of symptoms and severity of airflow limitation, further evaluation is warranted.

Figure 3. Pharmacologic treatment algorithms by Global Initiative for Chronic Obstructive Lung Disease grade. Green boxes and arrows indicate preferred treatment pathways. ICS = inhaled corticosteroid; LABA = long-acting β -agonist; LAMA = long-acting muscarinic antagonist.

improves arm strength and endurance and improves capacity for upper extremity activities (226).

Self-management education. An educational program should include smoking cessation, basic information about COPD, aspects of medical treatment (respiratory medications and inhalation devices), strategies to minimize dyspnea, advice about when to seek help, and possibly a discussion of advance directives and end-of-life issues.

End-of-life and palliative care. Patients should be informed that should they become critically ill, they or their family members may need to decide whether a course of intensive care is

likely to achieve their personal goals of care. Simple, structured conversations about these possible scenarios should be discussed while patients are in their stable state (227).

Nutritional support. For malnourished patients with COPD, nutritional supplementation is recommended.

Vaccination. Influenza vaccination is recommended for all patients with COPD. The pneumococcal vaccinations PCV13 and PPSV23 are recommended for all patients older than 65 years of age. PPSV23 is also recommended for younger patients with COPD with significant comorbid conditions, including chronic heart or lung disease (228).

Oxygen therapy. Long-term oxygen therapy is indicated for stable patients who have:

- PaO_2 at or below 7.3 kPa (55 mm Hg) or SaO_2 at or below 88%, with or without hypercapnia, confirmed twice over a 3-week period; or
- PaO_2 between 7.3 kPa (55 mm Hg) and 8.0 kPa (60 mm Hg), or SaO_2 of 88%, if there is evidence of pulmonary hypertension, peripheral edema suggesting congestive cardiac failure, or polycythemia (hematocrit $> 55\%$).

Ventilatory support. Noninvasive ventilation (NIV) is occasionally used in patients with stable very severe COPD.

NIV may be considered in a selected group of patients, particularly those with pronounced daytime hypercapnia and recent hospitalization, although contradictory evidence exists regarding its effectiveness (229). In patients with both COPD and OSA, continuous positive airway pressure is indicated (204).

Interventional bronchoscopy and surgery.

- In selected patients with heterogeneous or homogenous emphysema and significant hyperinflation refractory to optimized medical care, surgical or bronchoscopic modes of lung volume reduction (e.g., endobronchial one-way valves or lung coils) may be considered (230).
- In selected patients with a large bulla, surgical bullectomy may be considered.
- In selected patients with very severe COPD and without relevant contraindications, lung transplantation may be considered.

Choosing bronchoscopic lung reduction or LVRS to treat hyperinflation in a patient with emphysema depends on a number of factors, including: the extent and pattern of emphysema identified on HRCT, the presence of interlobar collateral ventilation measured by fissure integrity on HRCT or physiological assessment (endoscopic balloon occlusion and flow assessment), local proficiency in the

performance of the procedures, and patient and provider preferences. An algorithm depicting the various interventions on the basis of radiological and physiological features is shown in Figure 4.

Criteria for referral for lung transplantation include COPD with progressive disease, not a candidate for endoscopic or surgical lung volume reduction, BODE index of 5 to 6, P_{CO_2} greater than 50 mm Hg or 6.6 kPa and/or PaO_2 less than 60 mm Hg or 8 kPa, and FEV_1 less than 25% predicted (231). Recommended criteria for listing include one of the following: BODE index greater than 7, FEV_1 less than 15 to 20% predicted, three or more severe exacerbations during the preceding year, one severe exacerbation with acute hypercapnic respiratory failure, or moderate to severe pulmonary hypertension (231, 232).

Key points for the use of nonpharmacologic treatments are summarized in Table E9.

Monitoring and Follow-up

Routine follow-up of patients with COPD is essential. Symptoms, exacerbations, and objective measures of airflow limitation should be monitored to determine when to modify management and to identify any

complications and/or comorbidities that may develop. To adjust therapy appropriately as the disease progresses, each follow-up visit should include a discussion of the current therapeutic regimen. Symptoms that indicate worsening or development of another comorbid condition should be evaluated and treated.

Key Points:

- An exacerbation of COPD is an acute worsening of respiratory symptoms that results in additional therapy.
- Exacerbations can be precipitated by several factors. The most common causes are respiratory tract infections.
- The goal for treatment of exacerbations is to minimize the negative impact of the current exacerbation and to prevent subsequent events.
- Short-acting inhaled β_2 -agonists, with or without short-acting anticholinergics, are recommended as the initial bronchodilators to treat an acute exacerbation.
- Maintenance therapy with long-acting bronchodilators should be initiated as soon as possible before hospital discharge.
- Systemic corticosteroids improve lung function (FEV_1) and oxygenation and shorten recovery time and hospitalization duration.
- Antibiotics, when indicated, shorten recovery time and reduce the risk of early relapse, treatment failure, and hospitalization duration.
- Methylxanthines are not recommended owing to side effects.
- NIV should be the first mode of ventilation used to treat acute respiratory failure.
- After an exacerbation, appropriate measures for exacerbation prevention should be initiated.

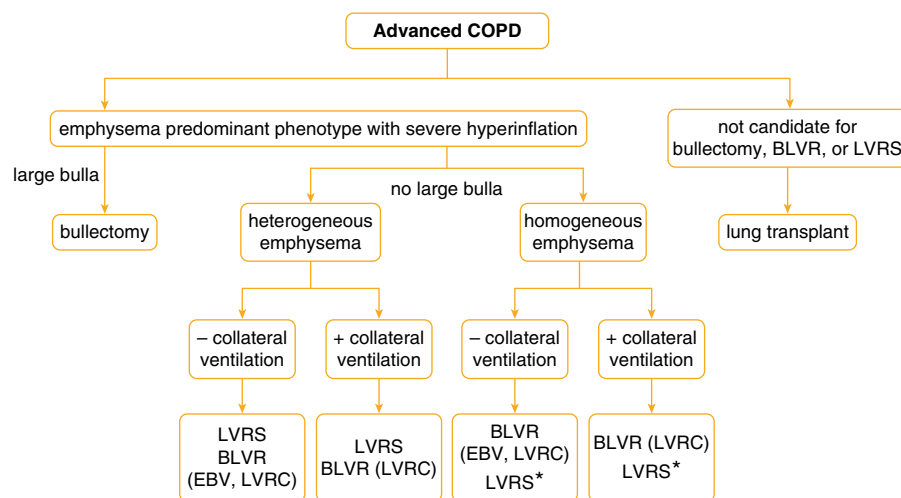


Figure 4. Interventional bronchoscopic and surgical treatments for chronic obstructive pulmonary disease (COPD). Overview of various therapies used to treat patients with COPD and emphysema worldwide. Note that all therapies are not approved for clinical care in all countries. In addition, the effects of BLVR on survival or other long term outcomes or comparison to LVRS are unknown. BLVR = bronchoscopic lung volume reduction; EBV = endobronchial valve; LVRC = lung volume reduction coil; LVRS = lung volume reduction surgery. *At some but not all centers.

Management of Exacerbations

Exacerbations are important events in the management of COPD because they negatively impact health status, rates of hospitalization and readmission, and disease progression (233, 234). COPD exacerbations

are complex events usually associated with increased airway inflammation, increased mucus production, and marked gas trapping. Increased dyspnea is the key symptom of an exacerbation. Other symptoms include increased sputum purulence and volume, together with increased cough and wheeze (235). As comorbidities are common in patients with COPD, exacerbations must be differentiated from acute coronary syndrome, worsening congestive heart failure, pulmonary embolism, and pneumonia.

COPD exacerbations are classified as:

- Mild (treated with short-acting bronchodilators only),
- Moderate (treated with short-acting bronchodilators plus antibiotics and/or oral corticosteroids), or
- Severe (patient requires hospitalization or visits the emergency room). Severe exacerbations may be associated with acute respiratory failure.

Exacerbations are mainly triggered by respiratory viral infections, although bacterial infections and environmental factors may also initiate and/or amplify these events (236).

Exacerbations can be associated with increased sputum production and, if purulent, increased bacteria may be found in the sputum (235, 237, 238). Some evidence supports the concept that eosinophils are increased in the airways, lung, and blood in a significant proportion of patients with COPD. Exacerbations associated with an increase in sputum or blood eosinophils may be more responsive to systemic steroids (239), although more prospective data are needed (239).

Symptoms usually last between 7 to 10 days during an exacerbation, but some events may last longer. At 8 weeks, 20% of patients have not recovered to their preexacerbation state (240). COPD exacerbations increase susceptibility to additional events (59, 241).

Patients with COPD susceptible to frequent exacerbations (defined as two or more exacerbations per year) have worse health status and morbidity than patients with less-frequent exacerbations (234). Other factors associated with an increased risk of acute exacerbations and/or severity of exacerbations include an increase in the ratio of the pulmonary artery to aorta cross-sectional dimension (i.e., ratio > 1) (242), a greater percentage of emphysema or airway wall thickness (243) measured by

chest CT imaging, and the presence of chronic bronchitis (244, 245).

Treatment Options

Treatment Setting

The goals of exacerbation treatment are to minimize the negative impact of the current exacerbation and to prevent the development of subsequent events (246). Depending on the severity of an exacerbation and/or the severity of the underlying disease, an exacerbation can be managed in either the outpatient or inpatient setting. More than 80% of exacerbations are managed on an outpatient basis with bronchodilators, corticosteroids, and antibiotics (59, 247, 248).

The indications for hospitalization during a COPD exacerbation are shown in Table E10. When patients with a COPD exacerbation come to the emergency department, they should be given supplemental oxygen and assessed to determine whether the exacerbation is life threatening and requires consideration for NIV and intensive care unit or respiratory unit hospitalization.

Long-term prognosis after hospitalization for COPD exacerbation is poor; the 5-year mortality rate is about 50% (249). Factors associated with poor outcomes include older age, lower BMI, comorbidities (e.g., cardiovascular disease or lung cancer), previous hospitalizations for COPD exacerbations, clinical severity of the index exacerbation, and need for long-term oxygen therapy at discharge (250, 251). Patients with a higher prevalence and severity of respiratory symptoms, poorer quality of life,

worse lung function, lower exercise capacity, lower lung density, and thickened bronchial walls on CT scan are at increased mortality risk after an acute exacerbation (252).

Key points for the management of all exacerbations are given in Table 8.

Pharmacologic Treatment

The most commonly used classes of medications for COPD exacerbations are bronchodilators, corticosteroids, and antibiotics.

Bronchodilators. Short-acting inhaled β_2 -agonists, with or without short-acting anticholinergics, are the initial bronchodilators recommended for acute treatment of exacerbations (253, 254). There are no significant differences in FEV₁ when using metered dose inhalers (with or without a spacer device) or nebulizers to deliver the agent (255), although the latter may be an easier delivery method for sicker patients. Intravenous methylxanthines are not recommended owing to side effects (256, 257).

Glucocorticoids. Systemic glucocorticoids in COPD exacerbations shorten recovery time and improve FEV₁. They also improve oxygenation (258–261), the risk of early relapse, treatment failure (262), and the length of hospitalization (258, 260, 263). A dose of 40 mg prednisone per day for 5 days is recommended (264). Therapy with oral prednisolone is equally effective to intravenous administration (265). Glucocorticoids may be less efficacious to treat exacerbations in patients with lower blood eosinophil levels (59, 239, 266).

Table 8. Key Points for the Management of Exacerbations

-
- Short-acting inhaled β_2 -agonists, with or without short-acting anticholinergics, are recommended as the initial bronchodilators to treat an acute exacerbation (**Evidence C**).
 - Systemic corticosteroids improve lung function (FEV₁) and oxygenation and shorten recovery time and hospitalization duration. Duration of therapy should not be more than 5–7 d (**Evidence A**).
 - Antibiotics, when indicated, can shorten recovery time and reduce the risk of early relapse, treatment failure, and hospitalization duration. Duration of therapy should be 5–7 d (**Evidence B**).
 - Methylxanthines are not recommended owing to increased side effect profiles (**Evidence B**).
 - NIV should be the first mode of ventilation used in patients with COPD with acute respiratory failure who have no absolute contraindication because it improves gas exchange, reduces work of breathing and the need for intubation, decreases hospitalization duration, and improves survival (**Evidence A**).
-

Definition of abbreviation: COPD = chronic obstructive pulmonary disease; NIV = noninvasive ventilation.

Antibiotics. The use of antibiotics in exacerbations remains controversial (267–269). Evidence supports the use of antibiotics in patients with exacerbations and increased sputum purulence (268, 269). One review reported that antibiotics reduce the risk of short-term mortality by 77%, treatment failure by 53%, and sputum purulence by 44% (270). Procalcitonin-guided antibiotic treatment may reduce antibiotic exposure and side effects with the same clinical efficacy (271, 272). A study in patients with exacerbations requiring mechanical ventilation (invasive or noninvasive) reported increased mortality and a higher incidence of secondary nosocomial pneumonia when antibiotics were not given (273). Antibiotics should be given to patients with acute exacerbations who have three cardinal symptoms: increase in dyspnea, sputum volume, and sputum purulence; have two of the cardinal symptoms, if increased purulence of sputum is one of the two symptoms; or require mechanical ventilation (invasive or noninvasive) (235, 236). The recommended length of antibiotic therapy is 5 to 7 days (274).

Antibiotic choice should be based on the local bacterial resistance pattern. Usual initial empirical treatment is an aminopenicillin with clavulanic acid, a macrolide, or a tetracycline. In patients with frequent exacerbations, severe airflow limitation (275, 276), and/or exacerbations requiring mechanical ventilation (277), cultures from sputum or other materials from the lung should be performed to identify the presence of resistant pathogens. Administration route depends on the patient's ability to eat and the pharmacokinetics of the antibiotic.

Respiratory Support

Oxygen therapy. Supplemental oxygen should be titrated to improve hypoxemia, with a target saturation of 88 to 92% (278). Once oxygen is started, blood gases should be checked to ensure satisfactory oxygenation without carbon dioxide retention and/or worsening acidosis.

Ventilatory support. Some patients require admission to the intensive care unit. Admission of patients with severe exacerbations to intermediate or special respiratory care units may be appropriate if

adequate personnel skills and equipment exist to manage acute respiratory failure.

NIV. NIV is preferred over invasive ventilation as the initial mode of ventilation to treat acute respiratory failure in patients hospitalized for acute exacerbations of COPD. NIV has been studied in RCTs showing a success rate of 80 to 85% (279–283). Mortality and intubation rates are reduced by NIV (279, 284–286).

Invasive mechanical ventilation. The indication for initiating invasive mechanical ventilation during an exacerbation includes failure of an initial trial of NIV (287). In patients who fail NIV as initial therapy and receive invasive ventilation as subsequent rescue therapy, morbidity, hospital length of stay, and mortality are greater (282).

Hospital Discharge and Follow-up

Lack of spirometric assessment and arterial blood gas analysis has been associated with rehospitalization and mortality (288). Mortality relates to patient age, the presence of acidotic respiratory failure, the need for ventilatory support, and comorbidities, including anxiety and depression (289).

The introduction of care bundles at hospital discharge to include education, optimization of medication, supervision and correction of inhaler technique, assessment and optimal management of comorbidities, early rehabilitation, telemonitoring, and continued patient contact have been investigated (290). There are insufficient data that they influence readmission rates, short-term mortality (288, 289, 291, 292), or cost-effectiveness (289).

Early follow-up (<30 d) after discharge should be undertaken when possible and has been related to fewer exacerbation-related readmissions (184, 293). Early follow-up permits a careful review of discharge therapy and an opportunity to make changes in therapy. Patients not attending early follow-up have increased 90-day mortality.

Additional follow-up at 3 months is recommended to ensure return to a stable state and review of patient's symptoms, lung function (by spirometry), and, when possible, the assessment of prognosis using multiple scoring systems such as BODE (293, 294). An assessment of the presence and management of comorbidities should also be undertaken (Table E11) (295).

Prevention of Exacerbations

After an acute exacerbation, measures for prevention of further exacerbations should be initiated (Table E12).

COPD and Comorbidities

Key Points:

- COPD often coexists with other diseases (comorbidities) that may significantly impact patient outcomes.
- The presence of comorbidities should not alter COPD treatment, and comorbidities should be treated per usual standards regardless of the presence of COPD.
- When COPD is part of a multimorbidity care plan, attention should be directed to ensure simplicity of treatment and minimize polypharmacy.

COPD often coexists with other diseases (comorbidities) that may have a significant impact on prognosis (63, 296–302). Some of these arise independently of COPD, whereas others may be causally related, either with shared risk factors or by one disease increasing the risk or compounding the severity of the other (303). Management of the patient with COPD must include identification and treatment of its comorbidities; the most common in COPD are outlined below.

Cardiovascular Disease

Heart failure. The prevalence of systolic or diastolic heart failure in patients with COPD ranges from 20 to 70% (304). Unrecognized heart failure may mimic or accompany acute exacerbations of COPD; 40% of patients with COPD who are mechanically ventilated because of hypercapnic respiratory failure have evidence of left ventricular dysfunction (305, 306). Treatment with β_1 -blockers improves survival in chronic heart failure and is recommended. Selective β_1 -blockers should be used (307).

Ischemic heart disease. There is an increased risk of myocardial damage in patients with concomitant ischemic

heart disease who have an acute exacerbation of COPD. Patients who demonstrate abnormal cardiac troponins are at an increased risk of adverse outcomes, including short-term (30 d) and long-term mortality (308).

Arrhythmias. Cardiac arrhythmias are common in COPD and vice versa. Atrial fibrillation is frequent and directly associated with FEV₁. Bronchodilators have been previously described as potentially proarrhythmic agents (309, 310); however, evidence suggests an overall acceptable safety profile for LABAs (311), anticholinergic drugs, and ICSs (101, 113, 133, 248, 312–316).

Peripheral vascular disease. In a large cohort of patients with COPD of all degrees of severity, 8.8% were diagnosed with peripheral artery disease (PAD), which was higher than the prevalence of 1.8% in control subjects without COPD (317). Patients with COPD with PAD reported a worse functional capacity and worse health status than those without PAD.

Hypertension. Hypertension is likely to be the most frequently occurring comorbidity in COPD and may have implications for prognosis (303, 318).

Osteoporosis

Osteoporosis is often associated with emphysema (319, 320), decreased BMI

(321), and low fat-free mass (322). Low bone mineral density and fractures are common in patients with COPD even after adjustment for steroid use, age, pack-years of smoking, current smoking, and exacerbations (323, 324). An association between inhaled corticosteroids and fractures has been found in pharmacoepidemiological studies. Systemic corticosteroids significantly increase the risk of osteoporosis.

Anxiety and Depression

Anxiety and depression are both associated with a poor prognosis (325, 326).

COPD and Lung Cancer

The association between emphysema and lung cancer is stronger than between airflow limitation and lung cancer (327–329). Increased age and greater smoking history further increase risk (330). Two studies of low-dose chest CT screening report improved survival in subjects aged 55 to 74 years, current smokers, or those who quit within the previous 15 years, with a smoking history of at least 30 pack-years (331, 332). Low-dose chest CT is now recommended in the United States for patients meeting these demographics; however, this is not a worldwide practice.

Metabolic Syndrome and Diabetes

Metabolic syndrome and diabetes are more frequent in COPD, and the latter is likely to affect prognosis (297). The prevalence of metabolic syndrome has been estimated to be more than 30% (333).

Gastroesophageal Reflux

Gastroesophageal reflux is an independent risk factor for exacerbations and is associated with worse health status (59, 334, 335).

Bronchiectasis

Bronchiectasis is associated with longer exacerbations (336) and increased mortality (295).

OSA

Patients with “overlap syndrome” (COPD and OSA) have a worse prognosis than patients with COPD or OSA alone. Patients with OSA and COPD with apneic events have more profound hypoxemia and more cardiac arrhythmias (337) and are more likely to develop daytime pulmonary hypertension (338, 339) than patients with just OSA or COPD alone. ■

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